

This writeup discusses discrepancy in quasi Monte-Carlo methods and introduces the Sobol sampling method.

1 Discrepancy

Definition 1.1 (Low discrepancy sequence). *A low discrepancy sequence is a sequence where all of its subsequences $X_1, \dots, X_N$ have low discrepancy.*

Roughly speaking, the discrepancy of a sequence is low if the proportion of points in the sequence falling into set B is close to proportional to the measure of B . A more precise definition is below.

Definition 1.2 (Discrepancy). *A sequence is “low-discrepancy” if the discrepancy of its first N terms satisfy:*

$$D_N^*(x_1, \dots, x_N) = O\left(\frac{(\log N)^s}{N}\right) \quad \text{as } N \rightarrow \infty.$$

Note: This is only a ‘customary’ statement, because for dimension $s \geq 2$, there remains an open problem: Does every infinite sequence $S \subset [0, 1]^s$ satisfy

$$D_N^*(S) = O\left(\frac{(\log N)^s}{N}\right) \quad \text{for infinite } N?$$

In this sense, the above bound for discrepancy may not be minimax, but it is the accepted norm when people refer to low discrepancy sequences.

Consider a point set $P = \{x_1, \dots, x_N\}$. Let:

- λ_s be the s -dim Lebesgue measure.
- $A(B; P)$ be the number of points in P in B , i.e., $A(B; P) = \sum_{i=1}^N \mathbf{1}(x_i \in B)$.
- $\mathcal{J}^*$ is the family of all subintervals of the form $\prod_{i=1}^s [0, u_i)$.

The star discrepancy $D_N^*(P) = D_N^*(x_1, \dots, x_N)$ of P is defined by

$$D_N^*(\mathcal{J}^*; P) = \sup_{B \in \mathcal{J}^*} \left| \frac{A(B; P)}{N} - \lambda_s(B) \right|.$$

One might be interested in bounding the error in doing the approximation of Monte Carlo methods. That is, what is the error

$$\int_{[0,1]^s} f(u) du \approx \frac{1}{N} \sum_{i=1}^N f(x_i)?$$

The Koksma–Hlawka inequality shows that this error is bounded by two terms:

1. A term depending only on f
2. A term depending on the discrepancy of the point set $\{x_1, \dots, x_N\}$

Theorem 1 (Koksma–Hlawka Inequality). *For any function f on $\bar{I}^s = [0, 1]^s$ of bounded variation $V(f) < \infty$ in the sense of Hardy and Krause, and any $x_1, \dots, x_N \in \bar{I}^s$, we have*

$$\left| \frac{1}{N} \sum_{n=1}^N f(x_n) - \int_{\bar{I}^s} f(u) du \right| \leq \underbrace{V(f)}_{\text{Analytic Complexity}} \cdot \underbrace{D_N^*(x_1, \dots, x_N)}_{\text{Geometric Uniformity}}.$$

We require the definition of $V(f)$:

Let $\Delta(f; J)$ be an alternating sum of values of f at vertices of J . The variation of f in the sense of Vitali is

$$V^{(s)}(f) = \sup_{\mathcal{P}} \sum_{J \in \mathcal{P}} |\Delta(f; J)| = \int_{\bar{I}^s} \left| \frac{\partial^s f}{\partial u_1 \dots \partial u_s} \right| du_1 \dots du_s$$

(if $\frac{\partial^s f}{\partial u_1 \dots \partial u_s}$ continuous).

And then the variation of f in the sense of Hardy and Krause is:

$$V(f) = \sum_{k=1}^s \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq s} V^{(k)}(f; i_1, \dots, i_k)$$

where $\{i_1, \dots, i_k\} \subseteq \{1, \dots, s\}$, ordered as $1 \leq i_1 < i_2 < \dots < i_k \leq s$ with f restricted to $\{(u_1, \dots, u_s) \in \bar{I}^s : u_j = 1 \ \forall j \neq i_1, \dots, i_k\}$.

The Koksma–Hlawka inequality motivates us to sample with as low star discrepancy $D_N^*(P)$ as possible.

The traditional Monte-Carlo sampling method achieves $\mathbb{E}[D_N^*] = O(N^{-1/2})$, which is considered slow. This can be shown relatively easily as below.

Suppose we want to evaluate $I = \int_{[0,1]^s} f(u) du$. By standard Monte Carlo, we draw $x_1, \dots, x_N \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}[0, 1]^s$ and $\hat{I}_N = \frac{1}{N} \sum_{n=1}^N f(x_n)$. Then,

$$\mathbb{E}[(\hat{I}_N - I)^2] = \text{Var}(\hat{I}_N) = \text{Var}\left(\frac{1}{N} \sum_{n=1}^N f(x_n)\right) = \frac{1}{N^2} N \sigma^2(f) = \frac{\sigma^2(f)}{N}.$$

The RMSE is then

$$\sqrt{\mathbb{E}[(\hat{I}_N - I)^2]} = \frac{\sigma(f)}{\sqrt{N}} = O(N^{-1/2}).$$

Using standard empirical process theory, we get $\mathbb{E}[D_N^*] = O(N^{-1/2})$.

2 Sobol Sequence

Next, we will discuss Sobol sampling, which has low discrepancy rate

$$D_N^*(S) \leq O\left(\frac{(\log N)^s}{N}\right) \quad \text{as } N \rightarrow \infty.$$

First, we require the following definitions.

Definition 2.1 (Elementary Interval in Base b). *Fix dim. $s \geq 1$ and integer $b \geq 2$. A subinterval E of I^s of the form*

$$E = \prod_{i=1}^s [a_i b^{-d_i}, (a_i + 1) b^{-d_i})$$

is an elementary interval in base b , where $a_i, d_i \in \mathbb{Z}$, $d_i \geq 0$, and $0 \leq a_i < b^{d_i}$ for $1 \leq i \leq s$.

Definition 2.2 $((t, m, s)$ -net). *Let $0 \leq t \leq m$ be integers. A (t, m, s) -net in base b is a point set P of b^m points in I^s such that $A(E; P) = b^t$ for every elementary interval E in base b with $\lambda_s(E) = b^{t-m}$.*

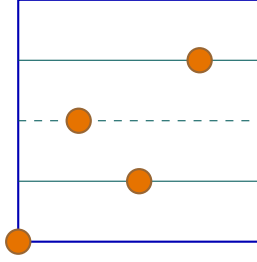
→ Intuitively, this is an s -dim point set so that every axis-aligned partition of the unit hypercube gets an exact allocation of points.

Example 1 $((0, 2, 2)$ -net in base $b = 2$). Let

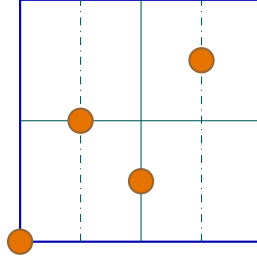
$$P = \left\{ (0, 0), \left(\frac{1}{2}, \frac{1}{4} \right), \left(\frac{1}{4}, \frac{1}{2} \right), \left(\frac{3}{4}, \frac{3}{4} \right) \right\}.$$

Here every elementary rectangle of area $\frac{1}{4}$ must contain exactly $2^0 = 1$ point.

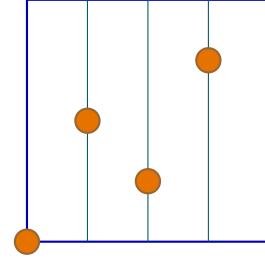
Below are all 3 ways to partition:



(a) $1 \times \frac{1}{4}$ strips



(b) $\frac{1}{2} \times \frac{1}{2}$ rectangles



(c) $\frac{1}{4} \times 1$ strips

Definition 2.3 $((t, s)$ -sequence). Let $t \geq 0$ be an integer. A sequence $x_0, x_1, \dots$ of points in I^s is a (t, s) -sequence in base b if, for all integers $k \geq 0$ and $m \geq t$, the point set consisting of x_n with $kb^m \leq n < (k+1)b^m$ is a (t, m, s) -net in base b .

Note: t is a parameter controlling the “uniformity”. A smaller t means more uniform.

The following theorem shows that the star discrepancy of the (t, s) -sequence matches the conjectured optimal asymptotic rate. Hence, a sampling sequence will be “good” if it is a (t, s) -sequence.

Theorem 2 (Niederreiter 1992, Thm 4.17). The star discrepancy of the first N terms of a (t, s) -sequence satisfies

$$D_N^*(S) \leq C(s, b) b^t \frac{(\log N)^s}{N} + O\left(b^t \frac{(\log N)^{s-1}}{N}\right) \quad \forall N \geq 2.$$

Now let us define the Sobol sequence, which is a (t, s) -sequence by construction. It was introduced by Ilya M. Sobol in 1967.

Construction of Sobol sequence

Let $I^s = [0, 1]^s$. A sequence $x_0, x_1, x_2, \dots \in I^s$ is a Sobol sequence if each coordinate of the point $x_n = (x_n^{(1)}, \dots, x_n^{(s)})$ is generated as follows:

1. **Binary expansion:** Write

$$n = \sum_{r=0}^{\infty} a_r(n) 2^r \quad \text{where } a_r(n) \in \{0, 1\}.$$

2. **Binary Digit expansion:** Write

$$x_n^{(i)} = \sum_{j=1}^{\infty} y_{nj}^{(i)} 2^{-j} \quad \text{where } y_{nj}^{(i)} \in \{0, 1\},$$

given by

$$y_{nj}^{(i)} = \sum_{r=0}^{\infty} c_{jr}^{(i)} a_r(n) \pmod{2}.$$

Here $c_{jr}^{(i)}$ are entries in the infinite generator matrix over $\mathbb{F}_2$.

What is the infinite generator matrix $C^{(i)} = (c_{jr}^{(i)})_{j \geq 1, r \geq 0}$? It is

$$C^{(i)} = \begin{bmatrix} | & | & \\ v_0^{(i)} & v_1^{(i)} & \dots \\ | & | & \end{bmatrix}$$

where $v_r^{(i)}$ are the direction numbers given by

$$v_r^{(i)} = \sum_{j=1}^{\infty} c_{j,r-1}^{(i)} 2^{-j} = \frac{m_r^{(i)}}{2^r}.$$

To generate the sequence $(m_r^{(i)})_{r \geq 1, \dots}$, write it as a monic primitive polynomial

$$P_i(x) = x^d + a_1 x^{d-1} + \dots + a_{d-1} x + 1$$

and choose d arbitrary odd integers $m_1^{(i)}, \dots, m_d^{(i)}$ so that each $m_j^{(i)} < 2^j$.

For $r > d$, compute

$$m_r^{(i)} = 2a_1 m_{r-1}^{(i)} \oplus 2^2 a_2 m_{r-2}^{(i)} \oplus \dots \oplus 2^d a_d m_{r-d}^{(i)} \oplus m_{r-d}^{(i)},$$

where $\oplus$ is the XOR bitwise operator.

Note: There is another algorithm, the Gray Code Construction (1979), which is faster.